NINE EQUIVALENT CONDITIONS FOR A SPECTRUM TO BE PROFINITE, HAUSDORFF OR TOTALLY DISCONNECTED

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Standard opens and quasi-compact opens of a spectrum. Used at the start of the counted proof (lines 4993-4995). Identifies quasi-compact opens with finite unions of standard opens.
- Closedness of a standard open containing a maximal point. Invoked in the counted proof at line 4996. Gives one of the implications between the topological conditions.

Source reference. Original lemma 00E8: sources/stacks-algebra/pinned-original-source.tex, source lines 3469–3469. Navigation pointer for \ref{lemma-quasi-compact} used inside the counted span (line 5012); the argument that consumes it is retained in the span.

Source reference. Original lemma 090M: sources/stacks-algebra/pinned-original-source.tex, source lines 4924–4924. Navigation pointer for \ref{lemma-spec-spectral} used inside the counted span (line 5017); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00EV: sources/stacks-algebra/pinned-original-source.tex, source lines 4956–4956. Navigation pointer for \ref{lemma-standard-open-containing-maximal-point} used inside the counted span (line 4997); the argument that consumes it is retained in the span.

Source reference. Original lemma 0905: sources/stacks-chow/tags.txt, source lines 11398–11398. Navigation pointer for \ref{topology-lemma-characterize-profinite-spectral} used inside the counted span (line 5018); the argument that consumes it is retained in the span.

Source reference. Original lemma 08ZY: sources/stacks-chow/tags.txt, source lines 11391–11391. Navigation pointer for \ref{topology-lemma-profinite} used inside the counted span (line 5013); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a ring. Let $X = \mathrm{Spec}(R)$ as a topological space. The following are equivalent*

(1) *X is profinite,*
(2) *X is Hausdorff,*
(3) *X is totally disconnected.*
(4) *every quasi-compact open of X is closed,*
(5) *there are no nontrivial inclusions between its prime ideals,*
(6) *every prime ideal is a maximal ideal,*
(7) *every prime ideal is minimal,*
(8) *every standard open $D(f) \subset X$ is closed, and*
(9) *add more here.*

Proof. First proof. It is clear that (5), (6), and (7) are equivalent. It is clear that (4) and (8) are equivalent as every quasi-compact open is a finite union of standard opens. The implication (7) $\Rightarrow$ (4) follows from Lemma 00EV. Assume (4) holds. Let $\mathfrak{p}, \mathfrak{p}'$ be distinct primes of R. Choose an $f \in \mathfrak{p}'$, $f \notin \mathfrak{p}$ (if needed switch $\mathfrak{p}$ with $\mathfrak{p}'$). Then $\mathfrak{p}' \notin D(f)$ and $\mathfrak{p} \in D(f)$. By (4) the open $D(f)$ is also closed. Hence $\mathfrak{p}$ and $\mathfrak{p}'$ are in disjoint open neighbourhoods whose union is X. Thus X is Hausdorff and totally disconnected. Thus (4) $\Rightarrow$ (2) and (3). If (3) holds then there cannot be any specializations between points of $\mathrm{Spec}(R)$ and we see that (5) holds. If X is Hausdorff then every point is closed, so (2) implies (6). Thus (2), (3), (4), (5), (6), (7) and (8) are equivalent. Any profinite space is Hausdorff, so (1) implies (2). If X satisfies (2) and (3), then X (being quasi-compact by Lemma 00E8) is profinite by Topology, Lemma 08ZY.

Second proof. Besides the equivalence of (4) and (8) this follows from Lemma 090M and purely topological facts, see Topology, Lemma 0905. □